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Laboratory of General Anatomy.
Course intended for first-year medical students.

Faculty of Medicine. Department of Medicine.
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INTRODUCTION TO HUMAN ANATOMY

PLAN

1. Definition
2. Branches of Anatomy
3. Method of Study
4. Anatomical Language
5. General Organization of the Human Body
6. Topographic Organization

Objectives:

- Define human anatomy.
- List the main branches of anatomy.
- Understand anatomical language.
- Know the different parts of the body.

1. Definition:

The term "anatomy" comes from the Greek "anatomia," meaning to cut through or dissect. Anatomy is the science of body structures, a fundamental science that studies the different organs of the human body. It deals with the description of organs (shape, size, color, weight) their relations, and their functions.

The study of human anatomy is of considerable practical interest; it is the basic science of medical studies, on which the study of physiology, clinical practice, and surgical technique relies.

2. Branches of Anatomy:

2.1. Descriptive Anatomy: the basic study of anatomy, studying separate organs, describing their shape, dimensions, weight, color, consistency, constitution, internal structure, etc.

2.2. Topographic or Regional Anatomy: study of the location and relations between organs within the same anatomical region. It is the basis of surgery.

2.3. Functional Anatomy: studies the relationship between the morphology and function of the organ.

- 2.4. Surface or Artistic Anatomy: study of the external forms of the human body.
- 2.5. Medical-Surgical Anatomy: anatomy applied to clinical practice or surgery.
- 2.6. Radiological Anatomy: study of morphology using imaging techniques (X-ray, CT, MRI, scintigraphy...).
- 2.7. Comparative Anatomy: compares the anatomy of humans and animals, or between humans.
- 2.8. Anthropological Anatomy: studies the external modifications of humans.
- 2.9. Developmental Anatomy: study of morphological transformations of the organism from fertilization to adulthood (embryology and growth).
- 2.10. Microscopic Anatomy: microscopic study of the structure of cells (cytology) and tissues (histology).
- 2.11. Pathological Anatomy: study of organ alterations caused by diseases.

3. Methods of Study:

There are several tools for studying anatomy: dissection, plastination, imaging (radiology, ultrasound, CT, MRI, arteriography, IVU, scintigraphy...), endoscopy (bronchoscopy, rectoscopy), surgery, microscope...

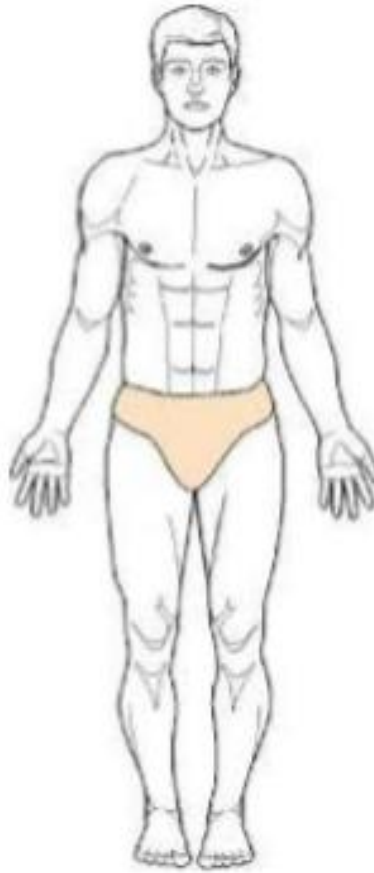
4. Anatomical Language:

To standardize terms and eliminate differences between countries, an international agreement (adopted in Paris in 1955) developed an international anatomical nomenclature in Latin, called Nomina Anatomica. It was updated in 1998 as Terminologia Anatomica (TA), also called the new nomenclature.

This nomenclature refers to a well-defined position of the human body, where three axes can be distinguished, which define three planes, as well as landmarks designating movement directions.

4.1. Reference Anatomical Position:

- Living subject standing upright.
- Horizontal gaze forward.
- Upper limbs hanging along the body.
- Palms facing forward.
- Feet flat on the ground, joined and parallel.

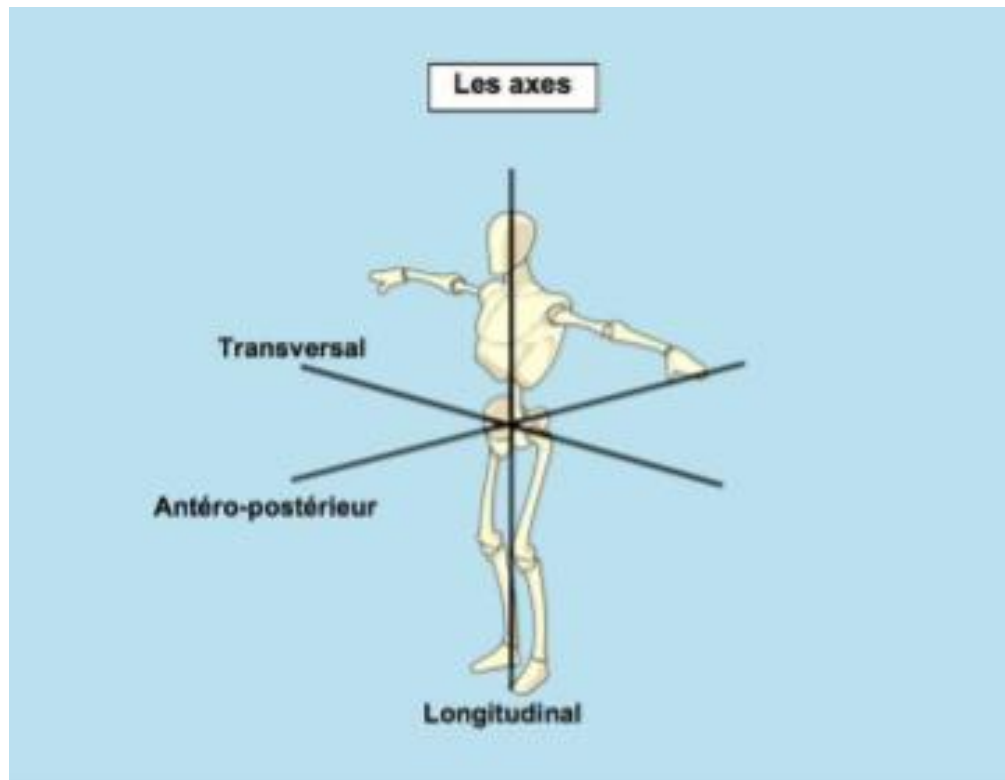


REFERENCE ANATOMICAL POSITION

4.2. Reference Axes:

An axis is an imaginary line, given by the intersection of two planes.

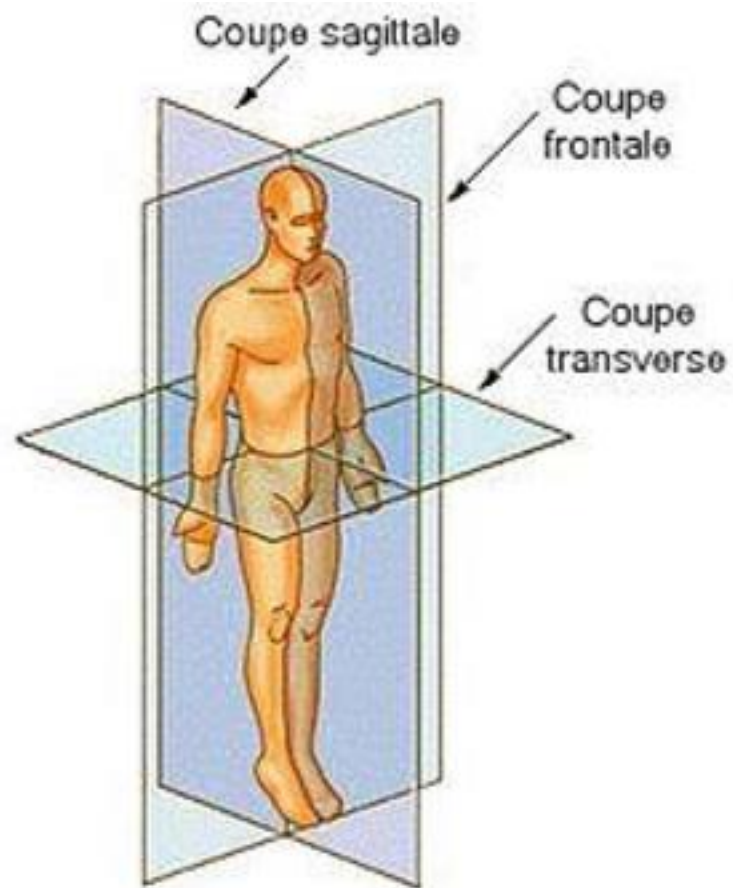
- Sagittal or anteroposterior axis: intersection of sagittal and transverse planes, horizontal, directed front to back.
- Frontal (transverse) axis: intersection of frontal and transverse planes, horizontal, directed side to side.
- Longitudinal axis: intersection of sagittal and frontal planes, vertical, cranio-caudal direction.
- Body axis: defined by the vertical passing through the vertex of the skull and the body's center of gravity (in the pelvis), with the subject in anatomical position.
- Hand axis: longitudinal axis passing through the 3rd finger.
- Foot axis: longitudinal axis passing through the 2nd toe.



4.3. Reference Planes:

The study of the human body is carried out in relation to three fundamental planes of space, which are perpendicular to one another.

- Sagittal plane: vertical, anteroposterior, divides the body into right and left parts.
 - * Median sagittal plane: passes through the body axis, divides into symmetrical right and left parts.
 - * Paramedian sagittal planes: parallel to the previous one.
- Frontal plane: vertical, parallel to the forehead, divides the body into anterior (ventral) and posterior (dorsal) parts.
 - * Coronal plane: frontal plane passing through the body axis.
- Transverse plane: horizontal, divides the body into cranial (superior) and caudal (inferior) parts.



REFERENCE PLANES



HAND AXIS AND FOOT AXIS

4.4. Dynamic Landmarks:

Movements of the body occur in a plane and around an axis. Three fundamental movements

- Flexion-Extension (sagittal plane, frontal axis)

flexion: reconciliation of segments

Extension: segments distance

- Abduction-Adduction (frontal plane, sagittal axis)

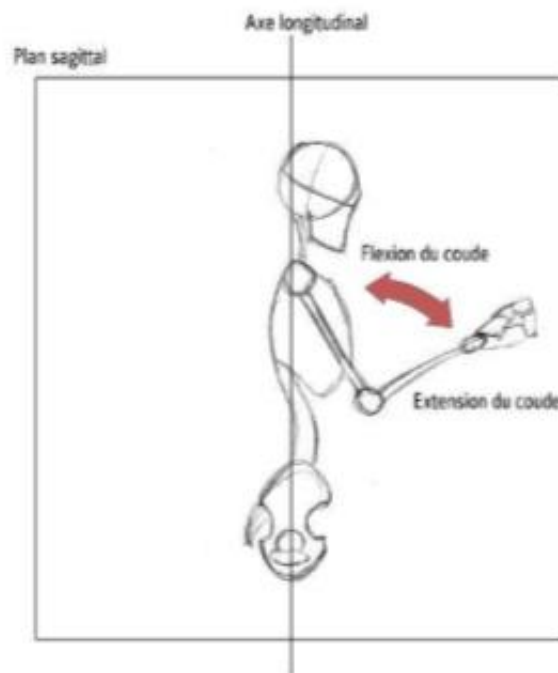
abduction: distance of segments from the median sagittal plane

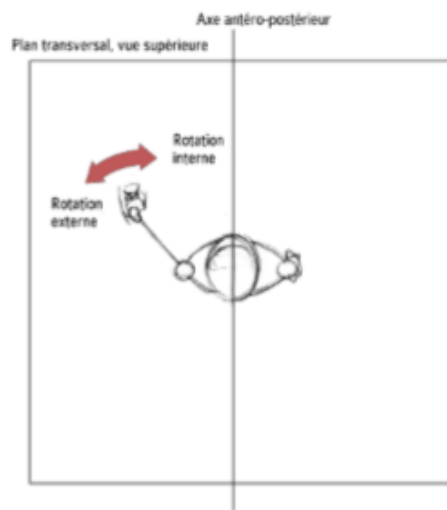
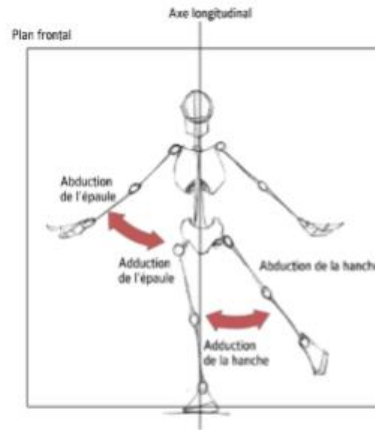
Adduction: reconciliation of segments.

- Internal and External Rotation (transverse plane, longitudinal axis)

External rotation: distance of the anterior face of the limb from the median sagittal plane.

Internal rotation: approximation of the anterior face of the member of the median sagittal plane.

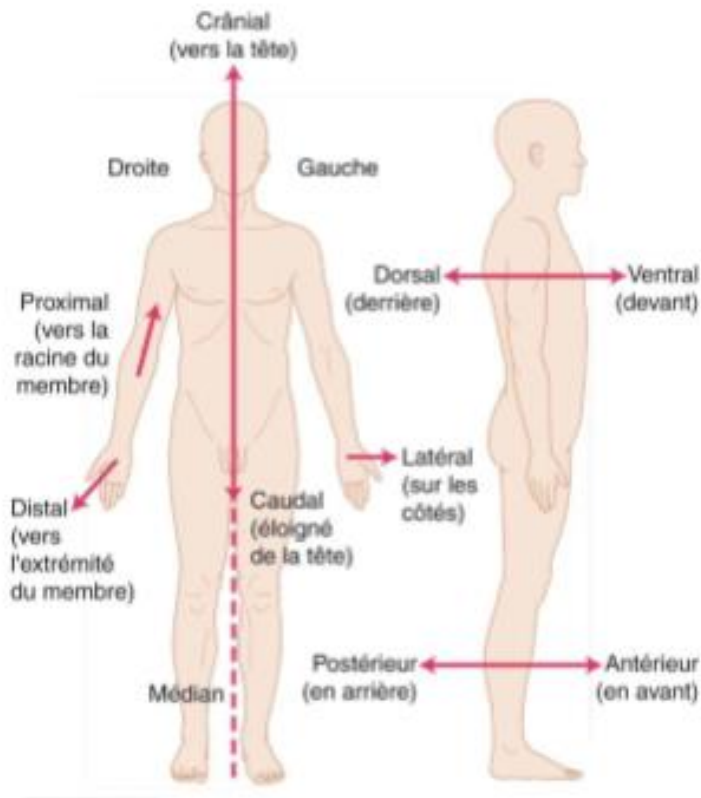




4.5. Localization Terms:

- Axial: located on the body axis.
- Abaxial: located away from the body axis.
- Internal: inside a cavity or viscus.
- External: outside a cavity or viscus.
- Cranial: near the upper end of the trunk.
- Caudal: near the lower end of the trunk.
- Proximal: near the root of a limb.
- Distal: near the end of a limb.
- Medial: near the median sagittal plane.
- Lateral: away from the median sagittal plane.
- Right: closer to the right side.
- Left: closer to the left side.
- Ipsilateral: belonging to the same side.
- Contralateral: belonging to opposite sides.
- Anterior or Ventral: near the anterior wall.

- Posterior or Dorsal: near the posterior wall.
- Superior and Inferior.
- Prefixes: supra = above, infra = below.



5. **General Organization of the Human Body:**

- 5.1. Cell: the morphological and functional unit of the organism.
- 5.2. Tissue: association of cells playing a similar role.
- 5.3. Organ: a set of varied tissues organized to fulfill a function.
- 5.4. System: a group of comparable organs structurally.
- 5.5. Apparatus: a set of different organs participating in the same function.

The human body consists of three groups of apparatus:

- Apparatuses of relational life: locomotor, nervous, and sensory apparatuses.
- Apparatuses of nutrition: digestive, circulatory, respiratory, urinary, and endocrine systems.
- Reproductive apparatus: male (testes, prostate, spermatic ducts) and female (ovaries, genital tracts).

6. Topographic Organization:

The body is composed of four parts:

- Head: contains initial segments of digestive and respiratory tracts, sensory apparatus, and a large part of the nervous system.
- Neck: passage region for visceral, vascular, and nervous elements.
- Trunk: thoracic cavity above and abdominopelvic cavity below.
- Limbs: mainly composed of bones, joints, muscles, vessels, and nerves.

Reference

1. Kamina P, Anatomie clinique, tome 1 anatomie générale membres, 4ème édition MALOINE. (Traduit en anglais).
2. Rouvière H. Anatomie humaine descriptive et topographique. Tome 1- tête et cou. Masson 10ème Ed 2ème, Paris(traduit en anglais).